Lecture 6: Four Fundamental Subspaces

MATH203 Applied Linear Algebra

Motivation: Connecting the Subspaces

Recall: The Preview from Lecture 5

At the end of Lecture 5, we discovered something beautiful:

Theorem (Preview)

$\text{Null}(A) \perp \text{Row}(A)$

Every vector in $\text{Null}(A)$ is orthogonal to every row of A.

Why? If $\mathbf{x} \in \text{Null}(A)$, then $A\mathbf{x} = \mathbf{0}$, which means:

$$\text{row}_i \cdot \mathbf{x} = 0 \quad \text{for all rows } i$$

Today's question: Is this an isolated phenomenon, or part of a bigger picture?

What We'll Discover Today

For any matrix A, there are **four fundamental subspaces**:

Subspace	**Lives in**	**Dimension**
$\mathrm{Col}(A)$	$\mathbb{R}^m$ (codomain)	r
$\mathrm{Null}(A)$	$\mathbb{R}^n$ (domain)	$n - r$
$\mathrm{Col}(A^T)$	$\mathbb{R}^n$ (domain)	r
$\mathrm{Null}(A^T)$	$\mathbb{R}^m$ (codomain)	$m - r$

Discovery: They pair up into **orthogonal complements**!

Roadmap for Today

Strategy: Build tools first, then reveal the full picture.

Part 1: How do subspaces behave under matrix products? (AB)

Part 2: What is transpose? (Flipping the ingredient table)

Part 3: The four fundamental subspaces and their orthogonality

Part 4: Dimension relationships

Goal: Understand the complete geometric structure of a matrix.

Subspaces of Matrix Products

Setup: Two-Stage Production

Imagine a **two-stage coffee shop**:

$$\begin{tikzcd} \boxed{\text{Raw materials}} \arrow[r, "B"] \arrow[rr, bend left, "AB"] & \boxed{\text{Semi-products}} \arrow[r, "A"] & \boxed{\text{Final products}} \end{tikzcd}$$

- B: raw materials $\rightarrow$ semi-products (e.g., roasted beans, brewed tea)
- A: semi-products $\rightarrow$ final products (e.g., latte, bubble tea)
- AB: raw materials $\rightarrow$ final products (combined process)

Question: How do the column spaces and null spaces of A, B, and AB relate?

Concrete Example: Two Matrices

Let's use actual matrices:

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad (3 \times 3)$$

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (2 \times 3)$$

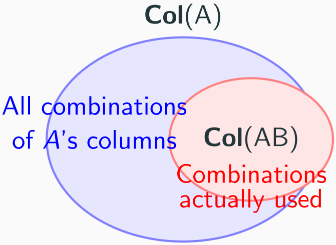
Product AB:

$$AB = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

We'll investigate: What are Col(A), Col(B), Col(AB)?

Visual: Column Space Containment

Key observation: Every column of AB is built from columns of A.



Why? Column j of AB equals A times column j of B.

So every column of AB is a **linear combination of A's columns**.

Theorem 1: Column Space of Product

Theorem

$\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$

The column space of AB is always contained in the column space of A.

Proof:

Write B with columns $\mathbf{b}_1, \ldots, \mathbf{b}_n$:

$$B = \begin{pmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Then AB has columns:

$$AB = \begin{pmatrix} | & | & & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Each $A\mathbf{b}_j$ is a linear combination of A's columns. $\square$

Why This Makes Sense: Column View of Matrix Product

Recall from Lecture 1: column j of AB equals $A\cdot$ (column j of B).

Example: If $\mathbf{b}_j = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and A has columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$:

$$A\mathbf{b}_j = 2\mathbf{a}_1 - 1\mathbf{a}_2 + 3\mathbf{a}_3$$

This is a **recipe** using A's columns as ingredients!

Conclusion: Every column of AB is cooked from A's columns.

Therefore $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$.

Verify with Our Example

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

Column space of AB:

$$\text{Col}(AB) = \text{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\}$$

Notice: $\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (dependent!)

So actually: $\text{Col}(AB) = \text{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$

Column space of A:

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$$

Indeed: $\text{Col}(AB) \subseteq \text{Col}(A)$ ✓ (Equality in this case!)

Theorem 2: Null Space Inclusion

Theorem

$\text{Null}(B) \subseteq \text{Null}(AB)$

The null space of B is always contained in the null space of AB.

Proof:

If $\mathbf{x} \in \text{Null}(B)$, then $B\mathbf{x} = \mathbf{0}$.

Therefore:

$$(AB)\mathbf{x} = A(B\mathbf{x}) = A(\mathbf{0}) = \mathbf{0}$$

So $\mathbf{x} \in \text{Null}(AB)$. $\square$

Why This Makes Sense: Linear Dependencies Persist

Recall from Lecture 5: $\mathbf{x} \in \text{Null}(B)$ records a **linear dependence** among columns of B.

Example: If $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \in \text{Null}(B)$, this says:

$$2\mathbf{b}_1 - 1\mathbf{b}_2 + 0\mathbf{b}_3 = \mathbf{0}$$

What happens when we apply A to both sides?

$$A(2\mathbf{b}_1 - 1\mathbf{b}_2) = 2(A\mathbf{b}_1) - 1(A\mathbf{b}_2) = 2(AB)_1 - 1(AB)_2 = \mathbf{0}$$

Conclusion: The **same linear dependence** holds for columns of AB!

Applying A preserves all existing linear relations.

Verify with Our Example

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Find Null(B):

By inspection, column 3 of B equals $2\times$ column 1 plus $1\times$ column 2:

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

So the dependence relation is: $-2\mathbf{b}_1 - 1\mathbf{b}_2 + 1\mathbf{b}_3 = \mathbf{0}$

$$\text{Null}(B) = \text{span}\left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Verify with Our Example (continued)

$$AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

Check: Does $\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \in \text{Null}(AB)$?

$$AB \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 - 1 + 5 \\ -2 + 0 + 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Yes! ✓

Find Null(AB) completely:

Column 3 of AB equals $2\times$ column 1 plus $1\times$ column 2:

$$\text{Null}(AB) = \text{span}\left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$

The Big Question

We've seen:

- $\text{Col}(AB) \subseteq \text{Col}(A)$ (always)
- $\text{Null}(B) \subseteq \text{Null}(AB)$ (always)

But: In our example, we had **equality** in both cases!

$$\textbf{When is Null}(AB) = \textbf{Null}(B) \textbf{ exactly?}$$

Meaning: When does multiplication by A **not create new dependencies**?

Theorem 3: The Equivalence

Theorem

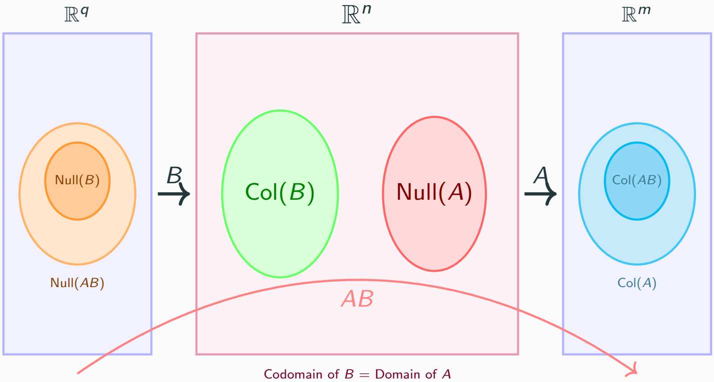
$$\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \quad \Longleftrightarrow \quad \text{Null}(AB) = \text{Null}(B)$$

Left side: Among all vectors B can produce, none (except $\mathbf{0}$) lands in Null(A).

Right side: All dependencies in AB were already in B (no new ones created).

These are **two perspectives on the same phenomenon**!

Visual: The Equivalence



Key correspondence: B maps Null(AB)$\setminus$ Null(B) onto Col(B)$\cap$ Null(A) $\setminus \{\mathbf{0}\}$

Therefore: Null(AB) $=$ Null(B) $\iff$ Col(B)$\cap$ Null(A) $= \{\mathbf{0}\}$

Proof Direction (a): $\Rightarrow$

Assume: $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

Goal: Show $\text{Null}(AB) = \text{Null}(B)$

We already know $\text{Null}(B) \subseteq \text{Null}(AB)$ (Theorem 2).

To show equality, prove the reverse inclusion.

Take any $\mathbf{x} \in \text{Null}(AB)$. Then $(AB)\mathbf{x} = \mathbf{0}$, so:

$$A(B\mathbf{x}) = \mathbf{0}$$

This means $B\mathbf{x} \in \text{Null}(A)$.

Proof Direction (a): $\Rightarrow$ (continued)

We have: $B\mathbf{x} \in \text{Null}(A)$.

But also: $B\mathbf{x} \in \text{Col}(B)$ (by definition of column space).

Therefore: $B\mathbf{x} \in \text{Null}(A) \cap \text{Col}(B)$.

By assumption, $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$, so:

$$B\mathbf{x} = \mathbf{0}$$

This means $\mathbf{x} \in \text{Null}(B)$.

We've shown $\text{Null}(AB) \subseteq \text{Null}(B)$. Combined with Theorem 2:

$$\text{Null}(AB) = \text{Null}(B)$$

$\square$

Proof Direction (b): $\Leftarrow$

Assume: $\text{Null}(AB) = \text{Null}(B)$

Goal: Show $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

Suppose $\mathbf{v} \in \text{Null}(A) \cap \text{Col}(B)$.

Since $\mathbf{v} \in \text{Null}(A)$: $A\mathbf{v} = \mathbf{0}$.

Since $\mathbf{v} \in \text{Col}(B)$: $\mathbf{v} = B\mathbf{x}$ for some $\mathbf{x}$.

Combining:

$$A(B\mathbf{x}) = A\mathbf{v} = \mathbf{0}$$

So $(AB)\mathbf{x} = \mathbf{0}$, meaning $\mathbf{x} \in \text{Null}(AB)$.

Proof Direction (b): $\Leftarrow$ (continued)

We have: $\mathbf{x} \in \mathrm{Null}(AB)$.

By hypothesis, $\mathrm{Null}(AB) = \mathrm{Null}(B)$, so:

$$\mathbf{x} \in \mathrm{Null}(B)$$

Therefore:

$$B\mathbf{x} = \mathbf{0}$$

But we had $\mathbf{v} = B\mathbf{x}$, so:

$$\mathbf{v} = \mathbf{0}$$

We've shown: any $\mathbf{v} \in \mathrm{Null}(A) \cap \mathrm{Col}(B)$ must be $\mathbf{0}$.

Therefore: $\mathrm{Null}(A) \cap \mathrm{Col}(B) = \{\mathbf{0}\}$ $\square$

Summary: Subspaces of Matrix Products

Theorem	Meaning
$\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$	Columns of AB built from A's columns
$\mathrm{Null}(B) \subseteq \mathrm{Null}(AB)$	Dependencies in B persist in AB
$\mathrm{Null}(A) \cap \mathrm{Col}(B) = \{\mathbf{0}\}$ $\Leftrightarrow$ $\mathrm{Null}(AB) = \mathrm{Null}(B)$	No new dependencies created

Key insight: Matrix multiplication has predictable effects on subspaces.

Next: We need a new tool to reveal deeper relationships — **transpose**.

Transpose: Flipping the Production Table

What Is Transpose? Flipping Perspective

In our ingredient table framework:

- **Rows** = raw materials (inputs)
- **Columns** = products (outputs)
- **Entry** A_{ij} = amount of material i needed for product j

Original question: Given a product, how much of each material do we need?

Flipped question: Given a material, which products use it?

This perspective flip is called **transpose**.

Concrete Example: Coffee Shop Ingredient Table

Original table A (materials $\to$ products):

	🥛	☕	🍵
🍃	0	0	2
🍋	0	0	1
🫘	0	2	4
🐄	1	0	1

$$= \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Reading:

- Column 🥛 : needs (0🍃, 0🍋, 0🫘, 1🐄)
- Row 🍃: distributed to products as $(0, 0, 2)$

Flipped Table: Transpose A^T

Flipped table A^T (products $\to$ materials):

	🍃	🍋	🫘	🐄
🥛	0	0	0	1
☕	0	0	2	0
🍵	2	1	4	1

$$= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 4 & 1 \end{pmatrix}$$

Reading:

- Row 🥛 : contains amounts (0🍃, 0🍋, 0🫘, 1🐄)
- Column 🍃: contributes to products as $(0, 0, 2)$

Notice: Rows of A become columns of A^T, and vice versa!

Visual: Flipping the Table

$$\begin{array}{c|ccc} & \text{🥛} & \text{☕} & \text{🍵} \\ \hline \text{🍃} & 0 & 0 & 2 \\ \text{🍋} & 0 & 0 & 1 \\ \text{🫘} & 0 & 2 & 4 \\ \text{🐄} & 1 & 0 & 1 \end{array} \quad \xrightarrow{\text{Flip}} \quad \begin{array}{c|cccc} & \text{🍃} & \text{🍋} & \text{🫘} & \text{🐄} \\ \hline \text{🥛} & 0 & 0 & 0 & 1 \\ \text{☕} & 0 & 0 & 2 & 0 \\ \text{🍵} & 2 & 1 & 4 & 1 \end{array}$$

A (materials = rows)

A^T (products = rows)

Key: The rows and columns swap roles!

Formal Definition

Definition (Transpose)

For an $m \times n$ matrix A, its **transpose** A^T is the $n \times m$ matrix:

$$(A^T)_{ij} = A_{ji}$$

Rows of A become columns of A^T, and columns of A become rows of A^T.

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Row 1 of A: $(1, 2, 3)$ becomes column 1 of A^T: $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

Property 1: Transpose of Product

Proposition

$$(AB)^T = B^T A^T$$

The transpose of a product reverses the order!

Proof (entry-by-entry):

Entry (i, j) of $(AB)^T$:

$$(AB)^T_{ij} = (AB)_{ji} = \sum_k A_{jk} B_{ki}$$

Entry (i, j) of $B^T A^T$:

$$(B^T A^T)_{ij} = \sum_k (B^T)_{ik} (A^T)_{kj} = \sum_k B_{ki} A_{jk}$$

Since multiplication of numbers commutes: $\sum_k A_{jk} B_{ki} = \sum_k B_{ki} A_{jk}$ $\square$

Why the Order Reverses: Production View

In a two-stage production $A \cdot B$:

- B: raw materials $\to$ semi-products
- A: semi-products $\to$ final products

Forward direction (AB): raw $\xrightarrow{B}$ semi $\xrightarrow{A}$ final

Flipped perspective (transpose):

- A^T: final product $\to$ which semi-products it uses
- B^T: semi-product $\to$ which raw materials it needs

Backward tracing: final $\xrightarrow{A^T}$ semi $\xrightarrow{B^T}$ raw

Hence $(AB)^T = B^T A^T$ — the order flips!

Property 2: Transpose of Inverse

Proposition

If A is invertible:

$$(A^{-1})^T = (A^T)^{-1}$$

Transpose and inverse commute.

Proof:

Start from $A \cdot A^{-1} = I$, transpose both sides:

$$(A \cdot A^{-1})^T = I^T = I$$

Using Property 1:

$$(A^{-1})^T \cdot A^T = I$$

Similarly, $(A^{-1} \cdot A)^T = I$ gives $A^T \cdot (A^{-1})^T = I$.

Therefore $(A^{-1})^T$ is the inverse of A^T. $\square$

Property 3: Transpose is Linear

Proposition

$$(A+B)^T = A^T + B^T$$

Transpose distributes over addition.

Proof (entry-by-entry):

$$\begin{aligned}(A+B)^T_{ij} &= (A+B)_{ji} \\ &= A_{ji} + B_{ji} \\ &= (A^T)_{ij} + (B^T)_{ij} \\ &= (A^T + B^T)_{ij}\end{aligned}$$

Therefore $(A+B)^T = A^T + B^T$. $\square$

Symmetric Matrices

Definition (Symmetric Matrix)

A matrix S is **symmetric** if $S^T = S$.

The ingredient table looks the same when flipped!

Example:

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$$

Flipping rows $\leftrightarrow$ columns gives the same matrix.

Notice: $S_{ij} = S_{ji}$ for all i, j (mirror symmetry across diagonal).

Theorem: AA^T and A^TA Are Always Symmetric

Theorem

For any matrix A:

1. AA^T is symmetric
2. A^TA is symmetric

Proof:

Using $(AB)^T = B^TA^T$:

$$(AA^T)^T = (A^T)^TA^T = AA^T$$

So AA^T is symmetric.

Similarly:

$$(A^TA)^T = A^T(A^T)^T = A^TA$$

So A^TA is symmetric. $\square$

Verify with Concrete Matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Compute AA^T:

$$AA^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

Check: $(AA^T)^T = AA^T$ ✓ (symmetric!)

Verify with Concrete Matrix (continued)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Compute $A^T A$:

$$A^T A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 35 & 44 \\ 44 & 56 \end{pmatrix}$$

Check: $(A^T A)^T = A^T A$ ✓ (symmetric!)

Change of Perspective

What we've learned so far:

- **Why** transpose exists: flipping the ingredient table perspective
- **Why** $(AB)^T = B^T A^T$: backward tracing reverses order
- **Why** AA^T is symmetric: algebraic consequence of transpose properties

From This Point Forward

We **forget about coffee shops** and work with abstract matrices A, B, C.

The key properties: $(AB)^T = B^T A^T$, symmetry of AA^T and $A^T A$.

These apply to **all matrices**, and we'll use them to discover deeper geometric relationships.

Next: The four fundamental subspaces and their beautiful orthogonality.

Four Fundamental Subspaces and Orthogonality

The Four Fundamental Subspaces

For any $m \times n$ matrix A, there are **four fundamental subspaces**:

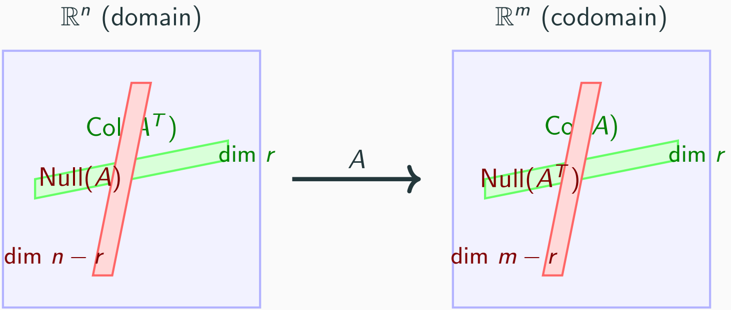
Subspace	Definition	Lives in	Dimension
$\text{Col}(A)$	$\{A\mathbf{x}\}$	$\mathbb{R}^m$	r
$\text{Null}(A)$	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	$\mathbb{R}^n$	$n - r$
$\text{Col}(A^T)$	$\{A^T\mathbf{y}\}$	$\mathbb{R}^n$	r
$\text{Null}(A^T)$	$\{\mathbf{y} : A^T\mathbf{y} = \mathbf{0}\}$	$\mathbb{R}^m$	$m - r$

where $r = \text{rank}(A)$.

Key observations:

- Two subspaces in $\mathbb{R}^n$ (domain): $\text{Null}(A)$ and $\text{Col}(A^T)$
- Two subspaces in $\mathbb{R}^m$ (codomain): $\text{Col}(A)$ and $\text{Null}(A^T)$

Visual: Two Pairs of Subspaces



Today's discovery: These pairs are **orthogonal** to each other!

Theorem 4: Col(A^T) $\perp$ Null(A)

Theorem

Every vector in $\mathrm{Col}(A^T)$ is orthogonal to every vector in $\mathrm{Null}(A)$.

Precisely: For $\mathbf{x} \in \mathrm{Null}(A)$ and $\mathbf{y} \in \mathrm{Col}(A^T)$,

$$\mathbf{y}^T\mathbf{x} = 0$$

Meaning: The row space and null space are perpendicular!

Proof (Inner Product View)

Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{y} \in \text{Col}(A^T)$.

By definition:

- $\mathbf{x} \in \text{Null}(A)$ means $A\mathbf{x} = \mathbf{0}$
- $\mathbf{y} \in \text{Col}(A^T)$ means $\mathbf{y} = A^T\mathbf{z}$ for some $\mathbf{z}$

Compute the inner product:

$$\begin{aligned} \mathbf{y}^T\mathbf{x} &= (A^T\mathbf{z})^T\mathbf{x} \\ &= \mathbf{z}^T(A^T)^T\mathbf{x} \\ &= \mathbf{z}^T A\mathbf{x} \end{aligned}$$

Since $\mathbf{x} \in \text{Null}(A)$, we have $A\mathbf{x} = \mathbf{0}$:

$$\mathbf{z}^T A\mathbf{x} = \mathbf{z}^T\mathbf{0} = 0$$

$\square$

Proof (Geometric View: Rows as Normal Vectors)

The equation $A\mathbf{x} = \mathbf{0}$ can be written row-by-row:

$$\begin{pmatrix} \text{—}\mathbf{r}_1^T\text{—} \\ \text{—}\mathbf{r}_2^T\text{—} \\ \vdots \\ \text{—}\mathbf{r}_m^T\text{—} \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{r}_1^T\mathbf{x} \\ \mathbf{r}_2^T\mathbf{x} \\ \vdots \\ \mathbf{r}_m^T\mathbf{x} \end{pmatrix} = \mathbf{0}$$

This means:

$$\mathbf{r}_1^T\mathbf{x} = 0, \quad \mathbf{r}_2^T\mathbf{x} = 0, \quad \ldots, \quad \mathbf{r}_m^T\mathbf{x} = 0$$

Interpretation: Each row $\mathbf{r}_i$ is **perpendicular** to $\mathbf{x}$.

Proof (Geometric View: continued)

We have: Each row $\mathbf{r}_i$ of A is perpendicular to every $\mathbf{x} \in \mathrm{Null}(A)$.

But: The rows of A are exactly the **columns of A^T**!

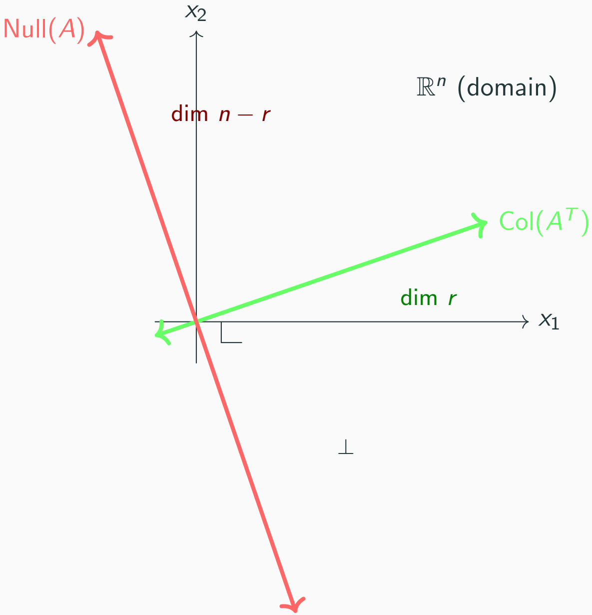
$$A^T = \begin{pmatrix} | & | & & | \\ \mathbf{r}_1 & \mathbf{r}_2 & \cdots & \mathbf{r}_m \\ | & | & & | \end{pmatrix}$$

Therefore:

$$\mathrm{Col}(A^T) = \mathrm{span}\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m\}$$

consists of vectors perpendicular to $\mathrm{Null}(A)$. $\square$

Visual: Orthogonality in $\mathbb{R}^n$



Concrete Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

Find Null(A):

$A\mathbf{x} = \mathbf{0}$ gives: $x_1 + 2x_2 + 3x_3 = 0$ (second row is $2\times$ first)

Basis for Null(A):

$$\text{Null}(A) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Find Col(A^T):

$$A^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$$

Verify Orthogonality

$$\text{Col}(A^T) = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}, \quad \text{Null}(A) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Check orthogonality:

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = 1(-2) + 2(1) + 3(0) = 0 \quad \checkmark$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}^T \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} = 1(-3) + 2(0) + 3(1) = 0 \quad \checkmark$$

Indeed, $\text{Col}(A^T) \perp \text{Null}(A)$!

Theorem 5: Col$(A) \perp$ Null(A^T)

Theorem

Every vector in Col(A) is orthogonal to every vector in Null(A^T).

Precisely: For $\mathbf{x} \in \text{Col}(A)$ and $\mathbf{y} \in \text{Null}(A^T)$,

$$\mathbf{y}^T\mathbf{x} = 0$$

Meaning: The column space and left null space are perpendicular!

Proof

Let $\mathbf{y} \in \mathrm{Null}(A^T)$ and $\mathbf{x} \in \mathrm{Col}(A)$.

By definition:

- $\mathbf{y} \in \mathrm{Null}(A^T)$ means $A^T\mathbf{y} = \mathbf{0}$
- $\mathbf{x} \in \mathrm{Col}(A)$ means $\mathbf{x} = A\mathbf{z}$ for some $\mathbf{z}$

Compute the inner product:

$$\begin{aligned} \mathbf{y}^T\mathbf{x} &= \mathbf{y}^T(A\mathbf{z}) \\ &= (A^T\mathbf{y})^T\mathbf{z} \end{aligned}$$

Since $A^T\mathbf{y} = \mathbf{0}$:

$$(A^T\mathbf{y})^T\mathbf{z} = \mathbf{0}^T\mathbf{z} = 0$$

$\square$

Symmetry of the Theorems

Beautiful duality:

- **Theorem 4**: $\mathrm{Col}(A^T) \perp \mathrm{Null}(A)$ (both in $\mathbb{R}^n$)
- **Theorem 5**: $\mathrm{Col}(A) \perp \mathrm{Null}(A^T)$ (both in $\mathbb{R}^m$)

These are **not two separate facts** — they're the **same theorem**!

Why? Apply Theorem 4 to the matrix A^T instead of A:

$$\mathrm{Col}((A^T)^T) \perp \mathrm{Null}(A^T) \quad \Rightarrow \quad \mathrm{Col}(A) \perp \mathrm{Null}(A^T)$$

which is exactly Theorem 5!

Deep insight: Transpose reveals a fundamental symmetry in linear algebra.

Concrete Example: The Dual Case

Using the same matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

Find Col(A):

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix} \right\} = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$$

Find Null(A^T):

$A^T\mathbf{y} = \mathbf{0}$ gives: $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{0}$

All equations reduce to: $y_1 + 2y_2 = 0$

$$\text{Null}(A^T) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$$

Concrete Example: Verify Orthogonality

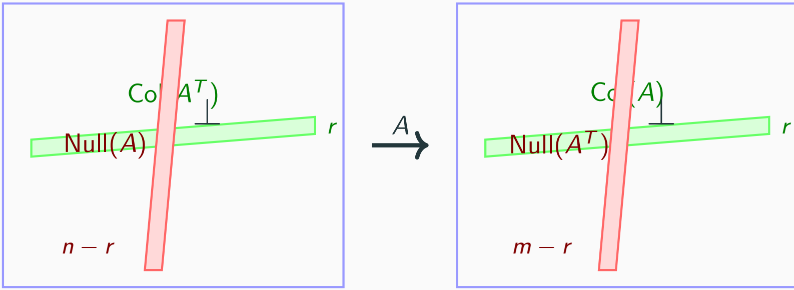
$$\mathrm{Col}(A) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}, \quad \mathrm{Null}(A^T) = \mathrm{span}\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$$

Check orthogonality:

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (-2)(1) + (1)(2) = -2 + 2 = 0 \quad \checkmark$$

Indeed, $\mathrm{Col}(A) \perp \mathrm{Null}(A^T)$!

Domain $\mathbb{R}^n$ **Codomain $\mathbb{R}^m$**



$\dim \text{Col}(A^T) + \dim \text{Null}(A) = r + (n - r) = n$ $\dim \text{Col}(A) + \dim \text{Null}(A^T) = r + (m$

Key: The orthogonal pairs completely fill their spaces!

Dimension Relationships

Theorem 6: Row Rank Equals Column Rank

Theorem

$$\mathrm{rank}(A) = \mathrm{rank}(A^T)$$

The dimension of $\mathrm{Col}(A)$ equals the dimension of $\mathrm{Col}(A^T)$.

In other words: **Column rank = Row rank**.

Proof idea (using cross-filling from Lecture 3):

Any matrix can be decomposed:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where U is $m \times r$ and V is $r \times n$.

Proof of Theorem 6 (continued)

$$A = UV$$

where:

- U is $m \times r$ with r linearly independent columns
- V is $r \times n$ with r linearly independent rows

Column rank:

$$\mathrm{Col}(A) = \mathrm{Col}(UV) = \mathrm{Col}(U)$$

So $\dim(\mathrm{Col}(A)) = r$.

Row rank:

The rows of $A = UV$ are linear combinations of the rows of V.

Since V has r linearly independent rows, $\dim(\mathrm{Col}(A^T)) = r$.

Therefore $\mathrm{rank}(A) = \mathrm{rank}(A^T) = r$. $\square$

Dimension Formulas

From the rank-nullity theorem (Lecture 5) and orthogonality:

Proposition

For an $m \times n$ matrix A with $\text{rank}(A) = r$:

In $\mathbb{R}^n$ (domain):

$$\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = r + (n - r) = n$$

In $\mathbb{R}^m$ (codomain):

$$\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = r + (m - r) = m$$

Meaning: The orthogonal pairs together **span their entire spaces**.

Summary

What We Achieved Today

1. **Subspaces of Matrix Products**:
   - $\text{Col}(AB) \subseteq \text{Col}(A)$ — columns of AB built from A's columns
   - $\text{Null}(B) \subseteq \text{Null}(AB)$ — dependencies persist
   - $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \Leftrightarrow \text{Null}(AB) = \text{Null}(B)$ — no new dependencies
2. **Transpose**:
   - Flipping the ingredient table perspective
   - $(AB)^T = B^T A^T$ — order reverses
   - AA^T and $A^T A$ always symmetric

What We Achieved Today (continued)

3. Four Fundamental Subspaces:

- $\text{Col}(A^T) \perp \text{Null}(A)$ in $\mathbb{R}^n$
- $\text{Col}(A) \perp \text{Null}(A^T)$ in $\mathbb{R}^m$
- These are dual statements — same theorem applied to A and A^T

4. Dimension Relationships:

- $\text{rank}(A) = \text{rank}(A^T)$ — column rank equals row rank
- $\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = n$
- $\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = m$

Key insight: The four fundamental subspaces form two orthogonal pairs that completely describe the geometry of a matrix.

Preview: Next Lecture

Lecture 7: Linear Transformations and Projections

Now that we understand the four fundamental subspaces and their orthogonality, we can study:

- What are linear transformations?
- What are projection operators? ($P^2 = P$)
- How do projections relate to the four subspaces?
- The geometric meaning of $\text{Col}(P)$ and $\text{Null}(P)$

Goal: Understand how to decompose vectors into orthogonal components.